NUMERICAL STUDY OF COBWEB CHANNEL WITH POROUS MEDIUM IN BATTERY COLD PLATE

PROJECT REPORT

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CERTIFICATE

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We, Midhun S, Giridhar Kumar M, Muhammed Basith N and Muhammed Mehran hereby declare that this project report entitled 'NUMERICAL STUDY OF COBWEB CHANNEL WITH POROUS MEDIUM IN BATTERY COLD PLATE' is the bonafide work of us carried out under the supervision of Prof. Faraz P Junaid, Assistant Professor, Department of Mechanical Engineering. We declare that, to the best of our knowledge, the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion to any other candidate. The content of this report is not being presented by any other student to this or any other University for the award of a degree.

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Finally, we thank our parents, friends, and all near and dear ones who directly and indirectly contributed to the successful completion of our project.

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ABSTRACT

Effective thermal management of lithium-ion batteries is critical for the safety, longevity, and performance of electric vehicles. This study numerically investigates the thermal and hydraulic performance of a novel cobweb-channel cold plate integrated with a highly conductive porous medium for Battery Thermal Management Systems (BTMS). Computational Fluid Dynamics (CFD) simulations were performed using ANSYS Fluent, following rigorous model validations of conjugate heat transfer and porous media flow against benchmark experimental data. The performance of the bionic cobweb cold plate was evaluated under a constant battery heat generation rate corresponding to a 3C discharge (32,357 W/m^3) across varying coolant inlet velocities (0.025 m/s to 0.15 m/s). To further augment heat dissipation, an aluminum metal with 95% porosity and a 0.2 mm pore diameter was introduced into the channels to maximize the effective heat transfer surface area.

The results demonstrate that the baseline cobweb geometry inherently achieves near-ideal thermal efficiency (>0.95) due to its superior radial-circumferential flow distribution. The integration of the metal successfully enhanced this performance, reducing the maximum battery-cold plate interface temperature by 0.32 K to 1.43 K across the tested velocities and achieving a peak cooling temperature of 302.68 K at 0.15 m/s. Alongside these thermal improvements, the hydraulic analysis quantified the expected increase in flow resistance associated with the porous structure, which measured 3.48 to 4.83 times that of the open channel. By comprehensively mapping these thermal-hydraulic characteristics, the study identifies an optimal operating velocity range of 0.10 m/s to 0.15 m/s. This range effectively balances maximum cooling performance with standard operational pumping requirements, providing practical design guidelines for next-generation, bionic-inspired EV cooling systems.

Keywords: *Battery Thermal Management System (BTMS), Cobweb Channel, Porous Media, Computational Fluid Dynamics (CFD), Conjugate Heat Transfer.*

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CHAPTER 1: INTRODUCTION

In recent years, electric vehicles have become very popular all over the world. This is because they are better for the environment and more efficient than traditional petrol or diesel cars. The most important part of an electric car is its battery pack. It acts like a fuel tank and engine combined. Most modern electric cars use Lithium-ion batteries. However, these Lithium-ion batteries have one major weakness: they generate a lot of heat. When we charge the battery or use it to drive the car, chemical reactions inside it produce heat. A battery has a comfortable temperature range, usually between 20°C to 40°C, where it works perfectly. If the battery gets too hot, its performance drops, it wears out much faster, and it can even get permanently damaged. In the most serious cases, this can lead to a dangerous situation called “the thermal runaway". This is a chain reaction where the battery heats itself up uncontrollably and can catch fire or explode; often the reason is that the battery got too hot and was not cooled properly.

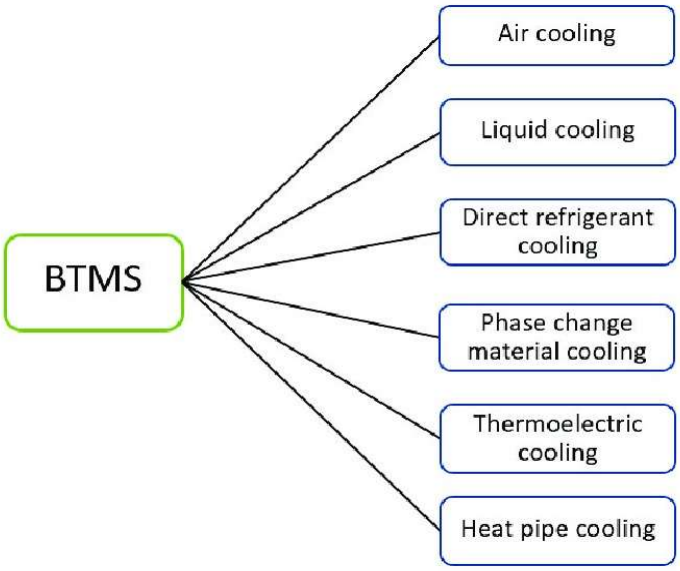


Figure 1.1 Classification of different battery thermal management system (BTMS) cooling technologies, Sukkam et al. (2022).

To prevent this, we need a very good cooling system for the battery, which is called a Battery Thermal Management System (BTMS). Think of it like the radiator in a regular car, but for the battery. There are a few ways to cool the battery. You can use air by blowing fans over it, but this is not very powerful for large, high-performance batteries. Another method uses special materials that melt to absorb heat, but this can be complex. The method that

works best for powerful batteries is liquid cooling. In this system, a special cool liquid is circulated around the battery to carry the heat away.

The main part of this liquid cooling system is the 'Cold Plate'. This is a solid metal plate, often made of aluminum, that is attached to the battery. Inside this plate, there are small channels or passages. The coolant liquid flows through these channels, absorbs the heat from the battery via the metal plate, and then carries that heat away to a radiator. The design of these internal channels is extremely important. A poor design can mean that some parts of the battery become very hot while other parts are cool. This uneven temperature is very bad for the battery's health and lifespan.

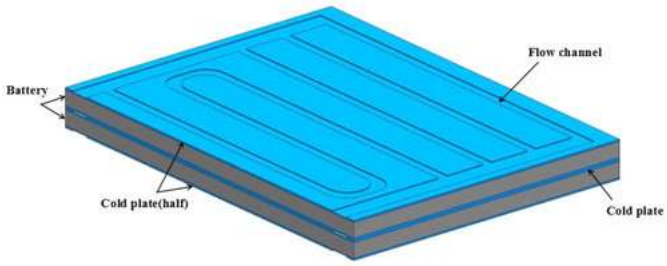


Figure 1.2 Schematic diagram of a liquid-cooled battery thermal management system (BTMS) using cold plates. Source: Dong et al. (2021).

Researchers are always looking for new and better channel designs. A common old design is the serpentine channel, which is like a long, winding river. The problem is that the coolant gets hotter as it travels, so the end of the channel is much hotter than the start. A new and very effective design is called the Cobweb Channel. This design is inspired by the pattern of a spider's web. It helps spread the coolant liquid evenly across the entire plate, so every part of the battery gets cooled similarly. This leads to a much more uniform temperature, which is the ideal situation.

We can make the cooling even better by using something called Porous Media inside these channels. Porous media is a material full of tiny, connected holes, like a metal sponge or foam. When we put this porous material inside the flow channel, it does two helpful things. First, it creates a huge amount of extra surface area inside the channel for the coolant to touch. More surface area means more heat can be transferred from the metal to the liquid. Second, it mixes and disturbs the flow of the coolant, which helps break up hot spots and makes the heat transfer more efficient.

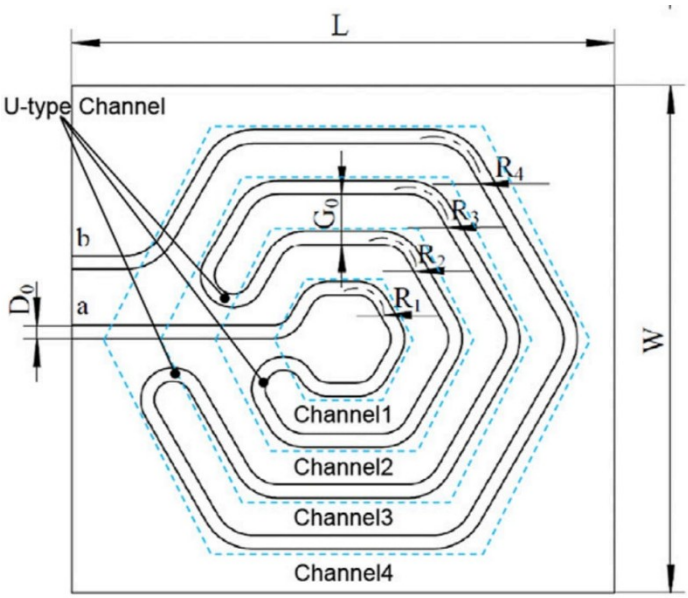


Figure 1.3, Diagram of cooling plate of cobweb-like channel structure.(Zhao et al., 2022)

Our project is a computer-based study where we combine these two excellent ideas. We are designing a cold plate that uses the novel cobweb channel, and we are filling these channels with a porous medium to boost its cooling power. We are using engineering software called Ansys Fluent to create a digital model of this cold plate and run simulations. These simulations allow us to observe coolant flow patterns and heat transfer behavior directly through computational analysis. This approach provides detailed insights into the early design stage.

The main goals for our project focus on two key performance areas. First, we aim to evaluate the thermal performance by analyzing how well our design cools the battery. We will examine both the maximum temperature reached and how even heat spreads across the battery surface. An ideal design will maintain low temperatures with minimal variation. Second, we will assess the hydraulic performance to understand the system's flow efficiency. While porous material enhances cooling, it also restricts coolant flow. If this restriction becomes too significant, the pump will need more power to circulate the coolant, making the system less efficient. Therefore, our objective is to achieve superior cooling performance while maintaining reasonable pumping power requirements.

CHAPTER 2: LITERATURE REVIEW

The thermal management of lithium-ion battery packs represents a critical challenge in the advancement of electric vehicle technology. Effective heat dissipation is essential for maintaining optimal operating temperatures, preventing thermal runaway, and ensuring long-term performance and safety. Recent research has increasingly focused on innovative cooling plate designs and the strategic integration of porous media to enhance thermal management efficiency. This literature review examines key contributions from recent studies that directly inform the present numerical investigation on cobweb channel configurations incorporating porous media in cold plate applications.

Zhao et al. (2022) introduced a novel cobweb-like channel cooling plate exchanger specifically designed for lithium-ion battery thermal management. Through comprehensive numerical research, they demonstrated that bionic designs inspired by natural spider web structures could substantially improve temperature distribution uniformity compared to conventional parallel or serpentine channel configurations. The cobweb geometry features radial channels emanating from a central hub with interconnecting circumferential pathways, promoting superior coolant distribution across the entire battery pack. This innovative topology effectively minimizes temperature gradients and eliminates localized hot spots that typically plague traditional channel designs. Their investigation revealed that asymmetric inlet and outlet configurations further enhanced flow distribution, achieving lower maximum temperatures and improved temperature uniformity indices. The study established critical design parameters including channel width ratios, branch angles, and hydraulic diameter optimization that directly influence thermal performance while maintaining acceptable pressure drop penalties.

Building upon the foundation of advanced channel geometries, Fu et al. (2025) conducted two complementary investigations exploring porous media integration in liquid cooling plates. Their first study examined cooling plates fully filled with porous medium, systematically investigating the effects of porosity, permeability, and porous material properties on thermal management performance. The porous structure fundamentally transforms the heat transfer mechanism by dramatically increasing the effective surface area available for convective heat exchange. The tortuous flow paths created within the porous matrix promote enhanced fluid mixing and eliminate boundary layer development, resulting

in significantly improved heat transfer coefficients. Their parametric analysis evaluated various metal foams and porous materials, demonstrating that optimized porosity selections could reduce maximum battery temperatures by 15-20% compared to conventional liquid cooling approaches while maintaining temperature uniformity within acceptable limits.

The second investigation by Fu et al. (2025) advanced this concept through multi-objective optimization of partially filled porous medium configurations. Recognizing that full porous filling incurs substantial pressure drop penalties and increased pumping power requirements, they strategically positioned porous inserts only in critical high-heat-flux regions. Employing Response Surface Methodology (RSM) to establish mathematical correlations between design variables and performance metrics, they generated comprehensive design spaces. The Non-dominated Sorting Genetic Algorithm II (NSGA-II) identified Pareto-optimal solutions balancing competing objectives of maximum temperature minimization, temperature uniformity enhancement, and pumping power reduction. Finally, the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) provided systematic ranking of optimal designs based on weighted performance criteria. This sophisticated multi-objective approach demonstrated that intelligent porous media placement could achieve 85-90% of the thermal benefits of full filling while reducing pressure drops by over 60%.

Wang et al. (2025) contributed valuable insights through multi-objective topology optimization of cold plates for battery thermal management in electric vehicles. Their approach employed density-based topology optimization algorithms to determine optimal material distribution within the design domain, effectively "growing" channel structures that naturally minimize thermal resistance. Unlike conventional designs, topology optimization enables exploration of unconventional configurations that human designers might not conceive. Their study balanced thermal performance objectives against manufacturing constraints including minimum feature size, structural connectivity, and fabrication feasibility. The resulting optimized geometries exhibited organic, fractal-like patterns that efficiently distributed coolant flow while minimizing material usage and system weight critical considerations for automotive applications.

Fazeli et al. (2011) conducted experimental and numerical investigations of heat transfer in miniature heat sinks utilizing silica nanofluid, providing foundational understanding of enhanced thermal transport mechanisms through nanoparticle suspension. Their work

demonstrated that nanofluids could significantly improve heat transfer coefficients in compact geometries, offering insights into advanced coolant formulations and their thermal-physical properties relevant to integrated cooling system design. The study established critical correlations between nanoparticle concentration, viscosity penalties, and overall thermal performance that inform coolant selection for battery thermal management applications.

Liu et al. (2018) investigated the effectiveness of nanofluids in mini-channel thermal management specifically for high-power lithium-ion batteries. Their research combined experimental validation with numerical modeling to assess how nanofluid coolants perform in realistic battery cooling scenarios. The study evaluated various nanofluid formulations under different flow conditions, demonstrating substantial improvements in heat removal capacity compared to conventional coolants. Their findings regarding pressure drop characteristics, thermal resistance reduction, and optimal flow rate selection provide essential design guidelines for implementing nanofluid-based cooling systems. This work bridges the gap between fundamental nanofluid research and practical battery thermal management applications, contributing methodologies for thermal-hydraulic.

Zolfagharnasab et al. (2022) conducted a comprehensive numerical investigation on porous-embedded shell-and-tube heat exchangers for waste heat recovery applications, focusing on the underlying mechanisms governing heat transfer enhancement rather than optimization alone. Their study demonstrated that incorporating porous media significantly improves thermal performance up to 60% increase in heat transfer by promoting two key mechanisms: enhanced zonal heat transfer through prolonged fluid–solid interaction and improved interfacial heat transfer via uniform temperature distribution at shell–tube interfaces None However, these gains are accompanied by substantial pressure drop penalties due to increased flow resistance within the porous structure. The work established that partially porous configurations offer an effective compromise, maintaining high thermal efficiency while reducing hydrodynamic losses.

Collectively, these investigations establish a comprehensive theoretical and practical framework for the current numerical study. The cobweb channel design offers inherent geometric advantages in flow distribution and pressure drop management, while porous media integration provides mechanisms for dramatic heat transfer enhancement through increased surface area and flow disruption. The application of multi-objective optimization

techniques enables systematic exploration of vast design parameter spaces to identify configurations that optimally balance thermal performance. By synthesizing these complementary approaches through detailed numerical simulation incorporating conjugate heat transfer, porous media modeling, and computational fluid dynamics, this research aims to develop next-generation battery thermal management solutions that meet the increasingly demanding requirements of high-energy-density battery systems in electric vehicle applications.

2.1. RESEARCH GAP

Existing literature demonstrates that porous media integration in liquid cooling plates significantly enhances thermal performance through increased surface area and improved convective heat transfer. However, previous investigations have predominantly focused on conventional channel geometries, particularly parallel and serpentine configurations. While these studies establish the thermal benefits of porous media insertion, they are limited by the inherent flow distribution characteristics of traditional channel designs, which often exhibit non-uniform coolant distribution and localized hot spot formation. Recent advances in bionic channel design, specifically cobweb-like geometries, have demonstrated superior temperature uniformity and reduced thermal gradients compared to serpentine layouts due to their radial-circumferential flow architecture that promotes more homogeneous coolant distribution across the cooling domain.

Despite the documented advantages of both porous media enhancement and cobweb channel topology independently, a critical research gap exists in understanding their combined performance characteristics. No existing study has systematically investigated the integration of porous media within cobweb-like channel configurations for battery thermal management applications. This gap is particularly significant because the synergistic effects of combining the enhanced heat transfer mechanisms of porous media with the superior flow distribution characteristics of cobweb geometry remain unexplored. Furthermore, the hydraulic implications of introducing porous resistance into the complex flow paths of cobweb channels, including pressure drop behavior and optimal porosity distribution strategies, have not been quantified. This study addresses this gap by conducting comprehensive numerical analysis of cobweb channel cold plates with porous media

integration, evaluating both thermal and hydraulic performance to establish design guidelines for this novel cooling configuration.

2.2. OBJECTIVES

The primary objective of this numerical study is to conduct comprehensive thermal performance analysis of the cobweb channel cold plate configuration integrated with porous media for lithium-ion battery thermal management. This investigation evaluates heat dissipation characteristics, maximum temperature distribution, and temperature uniformity across the battery module under various operating conditions. The analysis examines multiple porous media configurations, including different porosity values, permeability characteristics, and material properties, to determine their influence on thermal behavior. Particular emphasis is placed on minimizing temperature gradients to ensure homogeneous thermal conditions throughout the battery pack, thereby preventing localized thermal stress, capacity fade, and accelerated degradation. The thermal analysis quantifies the effectiveness of porous media insertion in reducing peak temperatures and improving overall thermal uniformity compared to conventional cooling plate designs without porous enhancement.

The secondary objective focuses on hydraulic performance analysis to assess the pressure drop penalty and pumping power requirements associated with porous media integration in the cobweb channel configuration. This evaluation examines the relationship between thermal enhancement and hydraulic resistance across varying porosity levels, flow rates, and channel geometries. The analysis quantifies pressure drop characteristics, flow distribution patterns, and energy consumption metrics to establish comprehensive performance maps. By examining the trade-offs between enhanced thermal performance and increased pumping power demands, this study aims to identify optimal design configurations that achieve superior thermal management while maintaining acceptable hydraulic efficiency. The findings will provide practical design guidelines for implementing cobweb-porous media cooling systems in electric vehicle battery applications, balancing thermal effectiveness with system-level energy considerations.

CHAPTER 3: METHODOLOGY

The overall objective of this research is to numerically investigate the thermal-hydraulic performance of a novel "Cobweb Channel" design integrated with porous media for application in battery cold plates. The project are divided majorly into three phases. In the first phase focuses on establishing confidence in the numerical models and techniques used in ANSYS Fluent. It is subdivided into two critical validation studies against experimental data:

Isothermal Fluid Flow Validation: This step validates the modelling of fluid flow over and through porous media. The experiment by Wu and Mirbod (2018), "Experimental analysis of the flow near the boundary of random porous media," is used as a first phase validation. The objective is to accurately replicate the interfacial slip velocity, flow penetration depth, and laminar velocity profiles, ensuring the porous zone model and solver settings are correctly configured.

Conjugate Heat Transfer Validation: This step validates the coupled fluid-solid heat transfer modelling essential for a thermal system like a cold plate. The study by Fazeli et al. (2012), "Experimental and numerical investigation of heat transfer in a miniature heat sink utilizing silica nanofluid," serves as the second validation. The goal is to verify the accuracy of thermal boundary conditions, heat flux application, and the prediction of key performance metrics such as thermal resistance and heat transfer coefficient.

After the successful validation, the project will proceed to the second phase where core design analysis has been done.A 3D model of the proposed cobweb channel will be designed, featuring a web-like flow path intended to enhance flow mixing and heat transfer surface area. Porous medium will be incorporated into specific regions of the cobweb channel. A comprehensive series of simulations will be conducted to analyse the effect of key parameters on performance, including porous medium properties, cobweb channel geometry, operating conditions etc. In the analysis stage we will analyse the thermal performance (heat dissipation efficiency, temperature uniformity) and hydraulic performance (pumping power requirement, pressure drop) will be evaluated. The design parameters will be optimized to achieve the best trade-off between maximizing heat transfer and minimizing pressure drop.

CHAPTER 4: NUMERICAL VALIDATIONS

4.1. INTRODUCTION

Computational Fluid Dynamics (CFD) is a branch of fluid mechanics that uses numerical methods and computational power to analyse and solve problems involving fluid flow, heat transfer, and associated phenomena. CFD is widely used across various industries, including aerospace, automotive, energy, and HVAC (Heating, Ventilation, and Air Conditioning), to optimize designs and improve system efficiency. By solving the Navier-Stokes equations, CFD allows engineers and researchers to simulate real-world fluid flow scenarios without the need for costly physical prototypes. Modern CFD software, such as ANSYS Fluent and OpenFOAM, integrates advanced turbulence models, multiphase flow analysis, and heat transfer simulations to provide highly accurate results.

4.2. NUMERICAL METHODS IN CFD

Computational Fluid Dynamics (CFD) relies on numerical methods to approximate solutions for complex fluid flow problems. Since the governing equations of fluid motion, known as the Navier-Stokes equations, are highly nonlinear and difficult to solve analytically, numerical techniques are used to break them down into simpler algebraic equations that can be solved computationally. The three primary numerical methods used in CFD are the Finite Difference Method (FDM), Finite Volume Method (FVM), and Finite Element Method (FEM), each with its own strengths and applications.

4.2.1 Finite Difference Method (FDM)

The Finite Difference Method (FDM) is one of the simplest numerical approaches, where derivatives in the governing equations are approximated using difference equations on a structured grid. It is particularly effective for solving problems with simple geometries and has a straightforward implementation. However, FDM struggles with complex geometries and irregular boundaries, making it less versatile for real-world CFD applications. Despite its simplicity, it is not commonly used in commercial CFD software due to its limitations in handling unstructured grids.

4.2.2 Finite Volume Method (FVM)

The Finite Volume Method (FVM) is the most widely used technique in CFD, particularly in software like ANSYS Fluent and OpenFOAM. This method divides the computational

domain into small control volumes, applying conservation laws in their integral form to ensure the accurate tracking of mass, momentum, and energy transfer. FVM is highly efficient for handling complex geometries and unstructured meshes, making it suitable for industrial applications such as aerodynamics, HVAC, and turbomachinery. While FVM ensures better conservation properties, it requires careful numerical treatment to maintain stability and accuracy in simulations.

4.2.3 Finite Element Method (FEM)

The Finite Element Method (FEM) is another powerful numerical approach, particularly useful for solving problems involving complex geometries and fluid-structure interactions. It divides the computational domain into smaller elements, using interpolation functions (shape functions) to approximate the solution. FEM is highly accurate and flexible, making it valuable for coupled physics simulations, such as those involving both fluid and solid mechanics. However, it is computationally expensive and more complex to implement in fluid flow problems, which is why it is more commonly used in structural analysis than in mainstream CFD applications.

Beyond these traditional methods, advanced techniques like Spectral Methods, Lattice Boltzmann Method (LBM), and Meshless Methods (e.g., Smoothed Particle Hydrodynamics - SPH) are gaining popularity for specialized applications. Spectral methods offer high accuracy for turbulence modelling, while LBM is useful for simulating complex flows at the microscopic scale. Meshless methods, such as SPH, are particularly effective for free-surface flows and multiphase problems.

Overall, the choice of numerical method in CFD depends on the complexity of the problem, required accuracy, computational resources, and the nature of the physical phenomena being studied. While FVM remains the most widely used approach in commercial CFD software, FEM and advanced numerical techniques continue to play crucial roles in specialized applications, contributing to the ongoing advancements in fluid dynamics simulations.

4.3 VALIDATION OF FLOW THROUGH POROUS MEDIUM

The initial and critical phase of this research involves the validation of the Computational Fluid Dynamics (CFD) methodology. This is essential to ensure that the numerical model accurately captures the complex physics of fluid flow interacting with a porous medium before it is applied to the novel cobweb channel design. The experimental study by Wu and

Mirbod (2018), titled "Experimental analysis of the flow near the boundary of random porous media," was selected for this validation. This paper provides a comprehensive and high-quality dataset that is ideally suited for our purpose for several key reasons. The authors investigated pressure-driven laminar flow by using within a simple rectangular channel, the bottom of which was coated with a soft, random porous material The porous media in the experimental analysis of this work, manufactured by Mountain Mist, have been made of 95% polyester and 5% silk. A configuration that shares the fundamental physics (interfacial slip, flow penetration, and Darcy type flow within the porous zone) with our proposed battery cold plate channel. Their work offers a detailed quantitative analysis, using Particle Image Velocimetry (PIV) to capture precise velocity profiles at the fluid-porous interface and within the porous medium itself, providing clear data for direct comparison.

To replicate their experiment numerically, our CFD model was built to mirror the exact experimental conditions. The geometry was created using the channel dimensions specified in the paper, The rectangular duct had a cross section of 25*42 mm^2 and a length of 90 cm. The porous layer heights of 5.5 mm, 9 mm, and 11 mm, as reported, were explicitly modeled. The working fluid was defined as a water-glycerin solution with a density of 1210 kg/m^3 and a dynamic viscosity of 0.07 kg/(m·s), matching the properties stated in the paper. The flow was simulated as laminar, consistent with the low Reynolds numbers (Re 2.5 and 7.5) maintained in the experiments. Crucially, the porous media properties, namely the porosity (ε = 0.95, 0.98, 0.99) and the derived permeability (K) values, were taken directly from the paper and implemented in the Fluent porous zone model. By replicating the experiment's geometry, fluid properties, flow regime, and porous media characteristics, we aim to achieve velocity profiles that match the experimental results, such as velocity fields obtained for various porosities and porous layer thicknesses through and above the porous media and comparison with the velocity profile obtained in an empty channel for Re = 2.5, mentioned in the paper. The successful replication of this validation case will confirm that our CFD model is correctly configured to simulate the coupled flow over and through porous media, thereby providing a validated foundation for the subsequent analysis of the more complex cobweb channel geometry for battery cooling applications.

Table 4.1, Test conditions and the related parameters of the present study. From Wu et al. (2018)

hp (mm)	ε	K(mm^2)
5.5	0.95	0.083
11	0.98	0.105
9	0.99	0.140

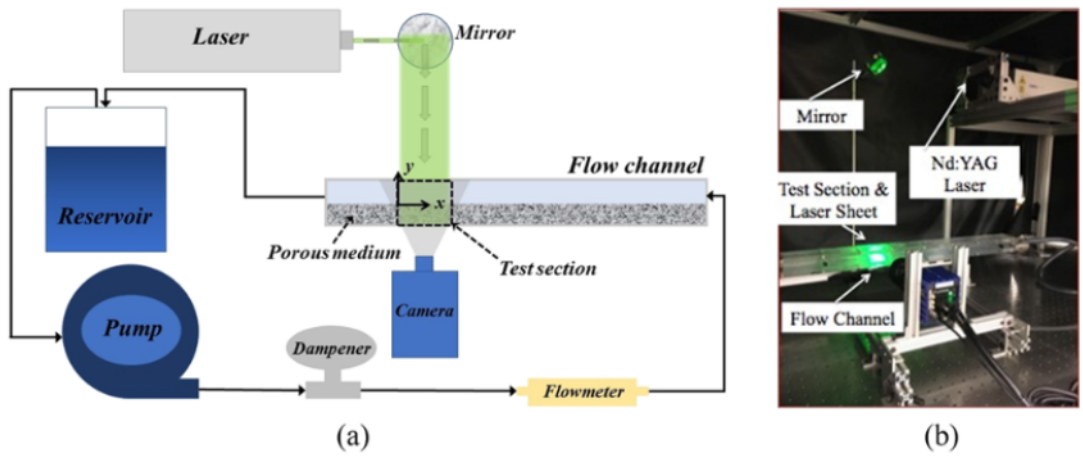


Figure 4.1, (a) Schematic diagram of the experimental setup. (b) Test section of the channel containing the random porous media covered the bottom wall of the channel, the location of the CCD camera and laser. Flow setup, including sample area of interest of PIV system. From Wu et al. (2018).

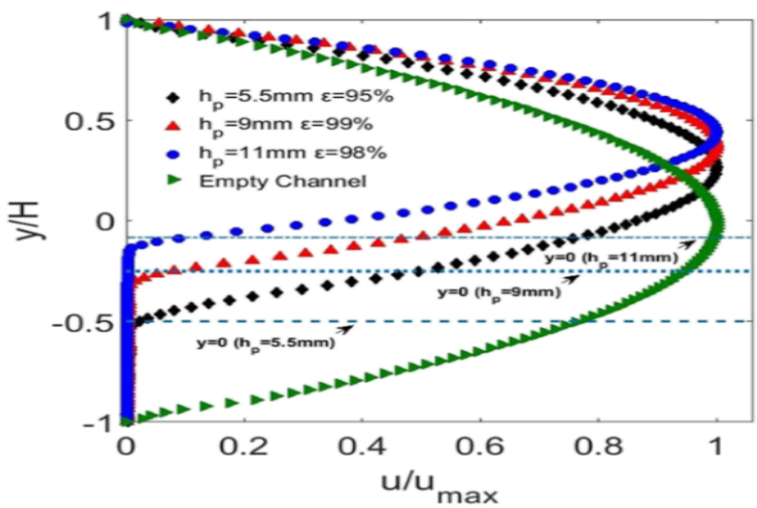


Figure 4.2, Velocity fields obtained for various porosities and porous layer thickness, Wu et al. (2018)

The CFD model was developed in ANSYS DesignModeler based on the experimental rectangular duct of Wu and Mirbod (900 × 42 × 25 mm) with a porous base layer (hp = 5.5–

11 mm), simplified to a 2D mid-plane (900 × 25 mm) to capture dominant streamwise–vertical flow behavior while reducing computational cost. A structured mesh with wall inflation layers was applied to resolve steep gradients near walls and the fluid–porous interface. The simulation in ANSYS Fluent used a pressure-based, steady-state, 2D planar solver with a laminar model (Re = 2.5–7.5). A water–glycerine fluid (ρ = 1210 kg/m³, μ = 0.07 kg/m·s) was defined, with separate fluid and porous zones; the latter employed a Darcy-based laminar porous model with porosity (0.95–0.99) and viscous resistance derived from permeability (e.g., K = 8.3×10^{-11} m²). Boundary conditions included a velocity inlet (0.048 m/s), pressure outlet (0 Pa), and no-slip walls. The SIMPLE algorithm with second-order upwind discretization was used, and the solution was inlet-initialized and iterated for 500 steps to ensure convergence

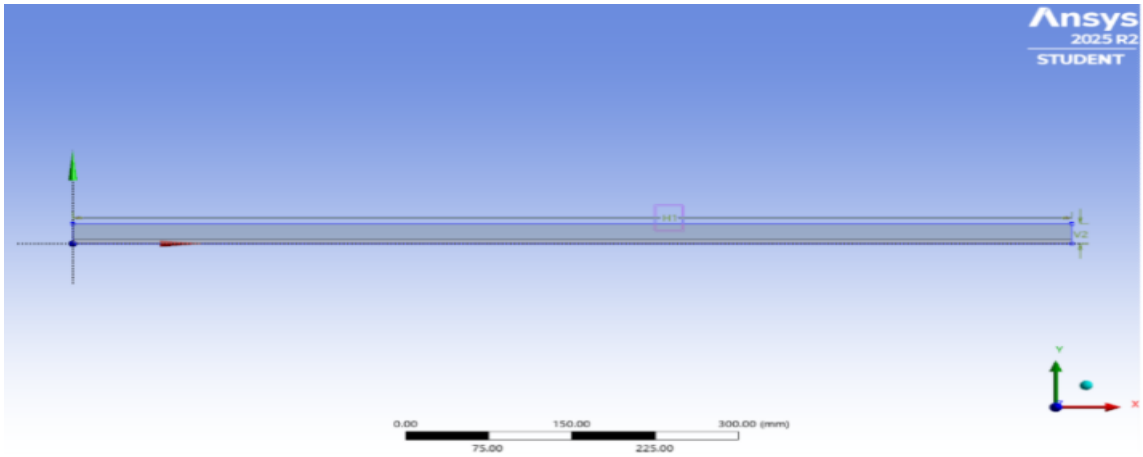
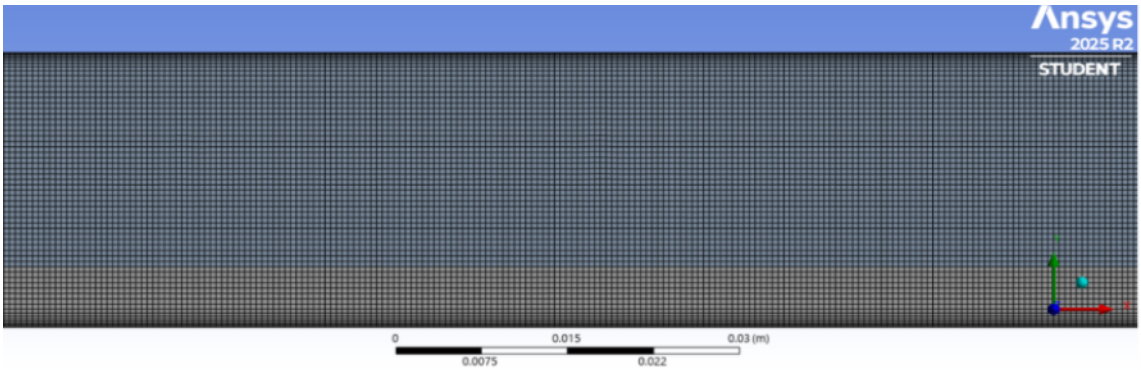


Figure 4.3, 2D Geometry of rectangular duct (H1=900mm, V2=25mm)



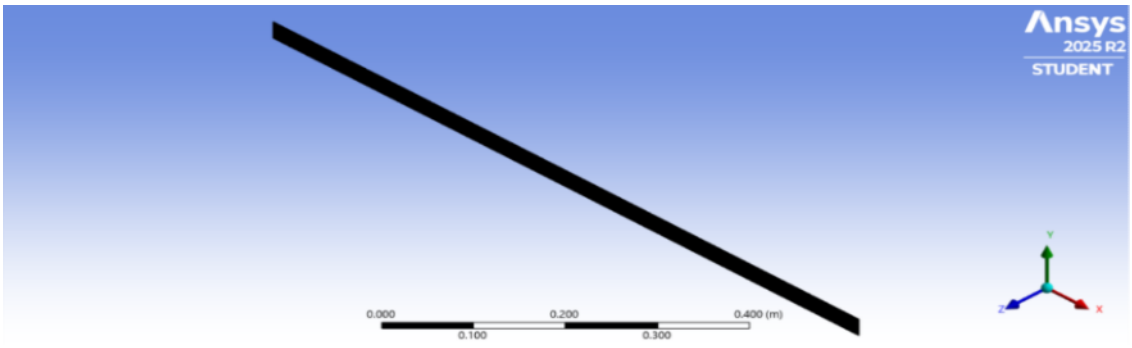


Figure 4.4, Generated mesh in the geometry

4.3.1 Result and Discussions

The CFD results were validated against the experimental velocity profiles of Wu and Mirbod, demonstrating non-classical behavior due to the porous layer, including finite slip velocity at the fluid–porous interface and momentum penetration into the porous medium with decay to Darcy velocity. Higher porosity and thickness increased slip velocity and reduced flow resistance, producing a flatter free-stream profile. The present simulation (hp = 5.5 mm, ε = 95%) accurately captured these effects, showing strong agreement in velocity distribution, interfacial slip, and decay trends, with minor discrepancies due to 2D simplification and idealized porous assumptions. Velocity contours further confirm a high-velocity core, clear interfacial momentum transfer, and rapid attenuation within the porous zone, validating the model's accuracy and robustness.

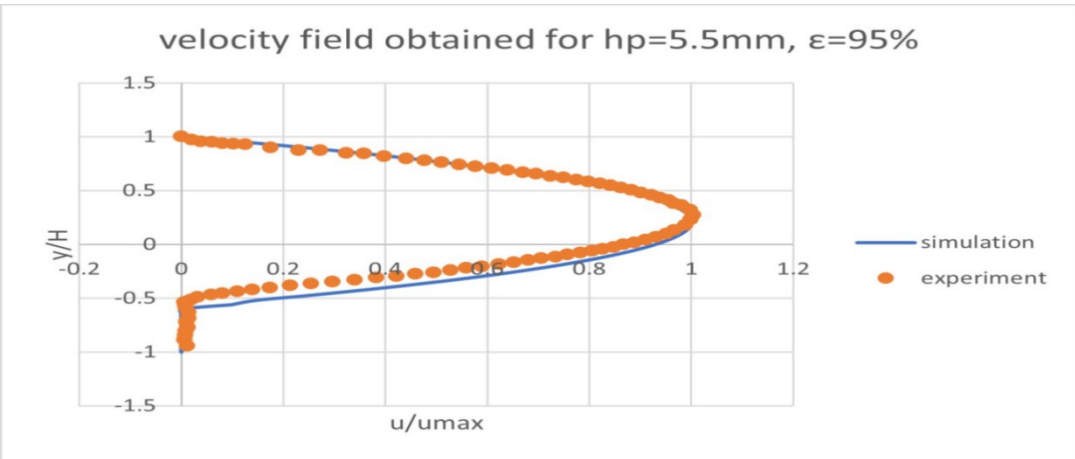


Figure 8: Velocity field obtained for porous height 5.5mm having porosity 95

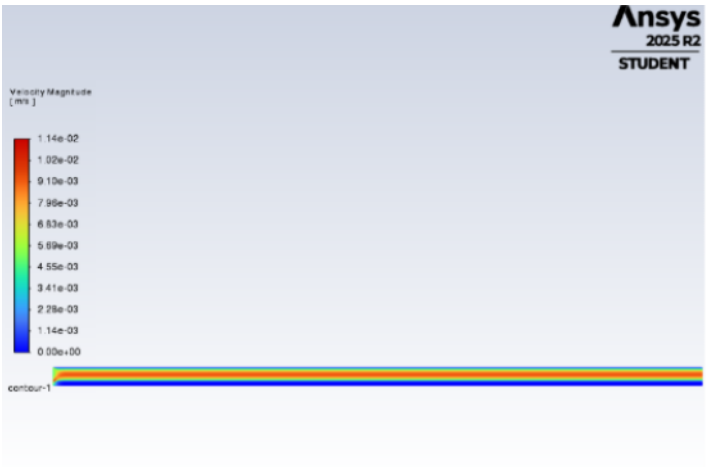


Figure 4.5, Velocity contour of flow through channel consist of hp=5.5 mm

4.4 VALIDATION OF COLD PLATE HEAT TRANSEFER

The initial and critical phase of this research involves the validation of the Computational Fluid Dynamics (CFD) methodology. This is essential to ensure that the numerical model accurately captures the complex physics of conjugate heat transfer and forced convection before it is applied to the novel cobweb channel cold plate design. The experimental and numerical study by Fazeli et al. (2012), titled "Experimental and numerical investigation of heat transfer in a miniature heat sink utilizing silica nanofluid," was selected for this validation. This paper provides a well-documented and rigorous dataset that is ideally suited for our purpose for several key reasons. The authors investigated a miniature aluminum heat sink with an array of circular channels under constant heat flux, cooled by SiO_2-water nanofluid. This configuration shares fundamental heat transfer mechanisms such as convective cooling, thermal resistance, and conjugate heat transfer with our proposed battery cold plate. Their work offers detailed experimental temperature distributions and heat transfer coefficients, which provide clear quantitative data for direct comparison with CFD results.

To replicate their study numerically, our CFD model was built to mirror the exact experimental conditions. The geometry was created using the dimensions specified in the paper: a square aluminum block of 40 mm × 40 mm × 10 mm, with circular channels of 4 mm diameter and 40 mm length. The working fluid was defined as water and SiO_2-water nanofluid with thermophysical properties. The flow was simulated as laminar, steady, and incompressible, consistent with the Reynolds number range of 400 to 2000 used in the experiments. A constant heat flux of 180 W/cm² was applied to the bottom surface, while all

other surfaces were treated as adiabatic. The SIMPLE algorithm was used for pressure-velocity coupling, and a second-order upwind scheme was employed for momentum and energy equations. A structured mesh with tri-elements was generated, and a grid independence study was conducted . By replicating the experiment's geometry, fluid properties, boundary conditions, and numerical methods, we aim to achieve temperature profiles and heat transfer coefficients that match the experimental results, such as the base plate temperature variation with flow rate and the heat transfer coefficient enhancement with Reynolds number. The successful replication of this validation case will confirm that our CFD model is correctly configured to simulate conjugate heat transfer in mini-channel cooling systems, thereby providing a validated foundation for the subsequent analysis of the more complex cobweb channel geometry for battery cooling applications.

The geometry for the validation study was created in ANSYS SpaceClaim based on the experimental work of Fazeli et al., using a simplified 3D model to capture key thermal-fluid behavior rather than exact geometric details. The model consisted of a 40 mm × 40 mm aluminum base plate with 8.5 mm thickness, containing five central circular channels of 4 mm diameter defined as fluid domains for conjugate heat transfer analysis, while omitting minor features such as manifolds to focus on conduction–convection interactions. Meshing was performed in ANSYS Meshing using a global element size of 5×10^{-4} m, with named selections for key regions and inflation layers at solid–fluid interfaces to resolve steep gradients, resulting in a hybrid mesh balancing accuracy and computational efficiency. The simulation in ANSYS Fluent employed a steady-state, laminar, pressure-based solver with the energy equation enabled; aluminum and water were assigned as solid and fluid materials respectively. Boundary conditions included a velocity inlet (0.071 m/s, 300 K), pressure outlet (0 Pa, 300 K backflow), constant heat flux of 100,000 W/m² at the base, and adiabatic external walls. The solution was initialized using hybrid initialization and iterated to convergence, ensuring stable residuals and accurate prediction of coupled heat transfer and flow behavior for validation against experimental data.

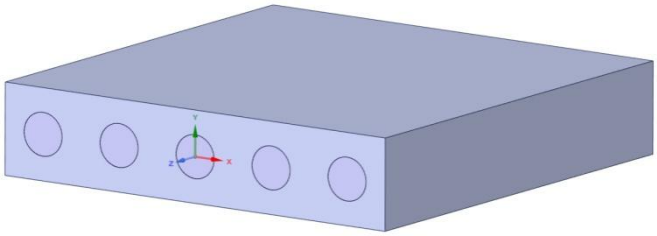


Figure 4.6, Geometry of the heatsink

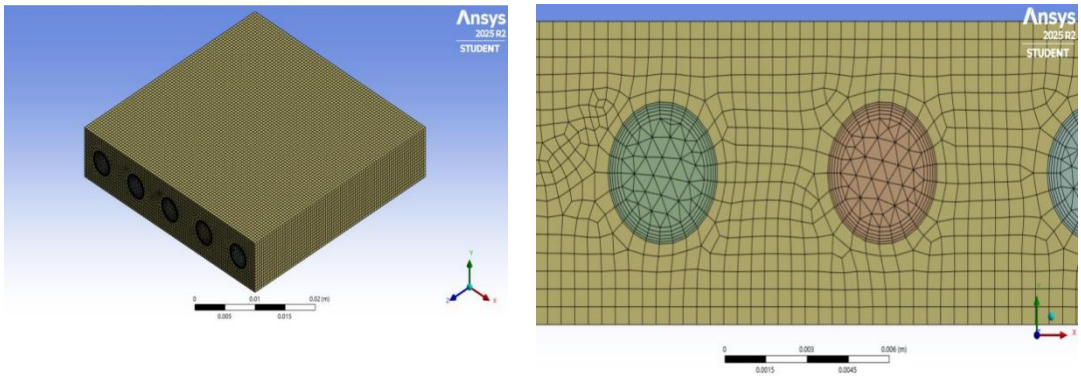


Figure 4.7, Generated mesh in the geometry

4.4.1 Results and Discussion

The thermal performance of the heat sink was evaluated by comparing CFD results with experimental data from Fazeli et al.. A line probe along the midsection of the heated base was used to extract temperature distribution, showing a steady increase in temperature along the flow direction due to continuous heat flux. Temperature contours further illustrated effective heat conduction through the aluminum and convective cooling within the channels, with higher temperatures near the base and lower values near fluid regions. The predicted temperature profile closely matched the experimental trend, confirming accurate capture of conjugate heat transfer behavior and validating the CFD setup, thereby establishing its reliability for further application to advanced designs such as the cobweb-channel cold plate.

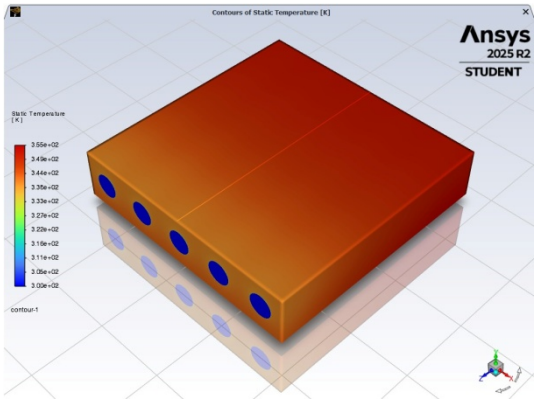


Figure 4.8, Contour: Static temperature

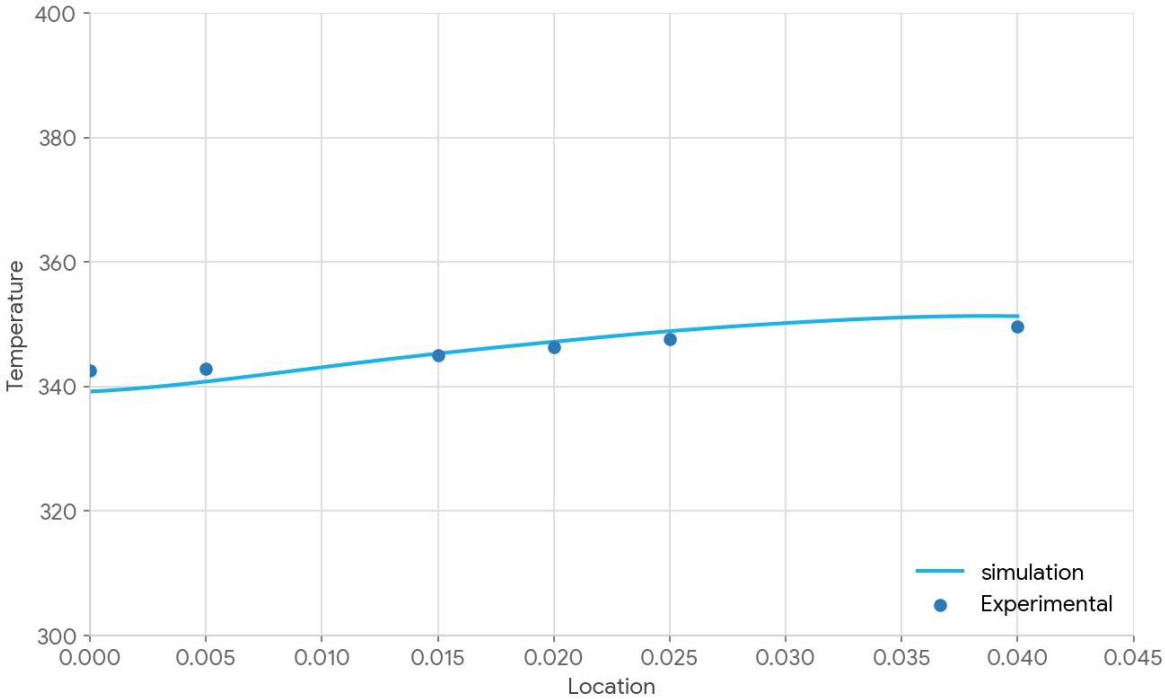


Figure 4.9, Base plate temperature variation with position

CHAPTER 5: NUMERICAL MODELING OF COBWEB CHANNEL

5.1 WITHOUT POROUS MEDIUM

Following the successful validation of the numerical methodology, the current phase of the project involves the design and assembly of a battery thermal management system featuring a novel cobweb-channel cold plate. The cold plate is intended to be sandwiched between two lithium-ion battery cells to maximize cooling contact area. Based on the battery specifications from Zhao et al. (2022), a rectangular battery with dimensions 148 mm × 27 mm × 91 mm was adopted. The cold plate was designed in SolidWorks with the same footprint (148 mm × 91 mm) to ensure full surface contact, and a thickness of 10 mm was selected to accommodate the internal cobweb-like channel network. The channels were patterned radially with a diameter of 3 mm, consistent with the parametric study range outlined in the reference. Both the cold plate and a half-battery model were created as separate components to leverage symmetry conditions, reducing computational expense. The half-battery was then mirrored and assembled on both sides of the cold plate in a sandwich configuration within SolidWorks. The assembly was imported into ANSYS SpaceClaim, where the fluid volumes inside the channels were explicitly defined, and shared topology was established at all interfacing surfaces between the battery, cold plate solid, and fluid domains to ensure proper conjugate heat transfer mapping. This modeling approach not only captures the physical arrangement of a practical battery module but also enables the use of symmetry boundary conditions to efficiently simulate conjugate heat transfer and fluid flow within the cobweb-channel cold plate under realistic thermal loads.

Following the geometry preparation in SpaceClaim, the assembly was discretized using ANSYS Meshing to enable accurate simulation of conjugate heat transfer and fluid flow. Key regions were first defined through named selections to facilitate boundary condition assignment and post-processing. These included the coolant inlet and outlet surfaces, the fluid body volume within the cobweb channels, the cold plate solid domain, the battery interfaces in contact with the cold plate, the fluid-solid interfaces along the channel walls, and the symmetry surfaces corresponding to the mid-plane of the full battery assembly. A structured meshing strategy was employed to balance computational efficiency with resolution requirements. A global element sizing of 3 mm was applied to all bodies to ensure consistent baseline resolution across the fluid and solid domains. To accurately resolve the steep thermal and velocity gradients near the channel walls, inflation layers were

added to the fluid boundaries interfacing with the cold plate. The inflation was configured using the smooth transition option with default growth rate and layer settings, ensuring a graded mesh that captures near-wall phenomena effectively. The final mesh was then generated, producing a hybrid grid that combines a uniform core mesh with boundary-conforming refinement at all critical interfaces. This approach ensures that the conjugate heat transfer between the battery, cold plate, and coolant is adequately resolved, supporting a reliable simulation of the thermal performance of the cobweb-channel cooling system.

The numerical simulation was configured within ANSYS Fluent to model the conjugate heat transfer and fluid flow within the battery–cold plate assembly. The energy equation was enabled, and the flow regime was set to laminar, consistent with the low Reynolds number operation of the cooling channels. Material properties were assigned to each domain: water was defined as the coolant with standard thermophysical properties, aluminum was assigned to the cold plate, and the battery domains were modeled using anisotropic thermal properties specifically, a through-plane conductivity of 1.1 W/m·K and in-plane conductivity of 30.8 W/m·K along with a density of 1923 kg/m³ and specific heat of 1535 J/kg·K, as reported in the reference study. In the cell zone conditions, water was assigned to the fluid channel volume, aluminum to the cold plate, and the battery material to each battery half-domain. A volumetric heat source of 32,357 W/m³ was activated within the battery zones to represent heat generation under a 3C discharge rate, matching the experimental condition specified in the paper. Boundary conditions were applied as follows: a velocity inlet of 0.05 m/s at a temperature of 300 K, a pressure outlet with 0 Pa gauge pressure and a backflow temperature of 300 K, and adiabatic conditions on all external walls to isolate the cooling effect to the cold plate. The solution was initialized using standard hybrid initialization and iterated until convergence in continuity, momentum, and energy residuals was achieved, establishing a steady-state baseline for thermal performance evaluation. This baseline configuration, without the inclusion of porous media, serves as the reference case against which the enhanced thermal performance of the porous-medium-integrated cold plate is evaluated.

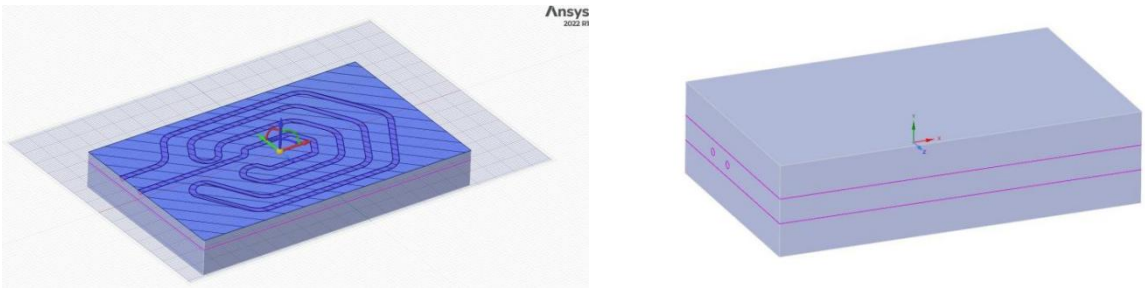


Figure 5.1, Geometry of Cobweb model

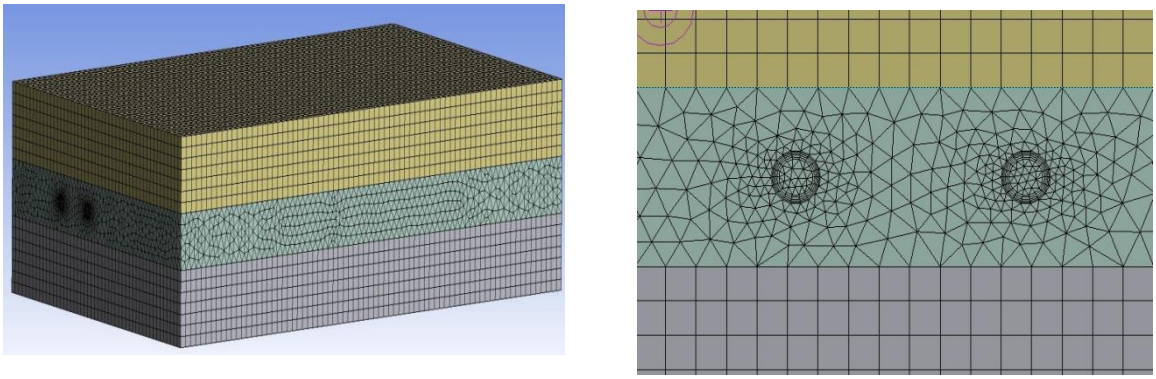


Figure 5.2, Generated mesh in the geometry

5.2 GRID INDEPENDENCE STUDY

A systematic grid independence study was conducted to determine the optimal mesh resolution that balances computational accuracy and cost. Three mesh configurations were generated by progressively refining the body sizing element size while maintaining consistent inflation parameters across all cases, including smooth transition inflation with 5 layers and a growth rate of 1.2 applied to the fluid channel walls. The coarse mesh with a body sizing of 0.75 mm produced 4,672,670 cells and yielded an interface temperature of 308.10 K. The medium mesh with a body sizing of 0.50 mm produced 12,375,502 cells and yielded an interface temperature of 309.57 K. The fine mesh with a body sizing of 0.20 mm produced 20,078,334 cells and yielded an interface temperature of 309.57 K. The results indicate that refining the mesh from the coarse to the medium configuration resulted in a temperature change of 1.47 K, while further refinement from the medium to the fine configuration produced no change in the interface temperature, confirming that grid independence was achieved at the medium mesh level. Therefore, the medium mesh with a body sizing of 0.50 mm and 12,375,502 cells was selected for all subsequent simulations in

this study, as it provides grid-independent results while maintaining reasonable computational efficiency.

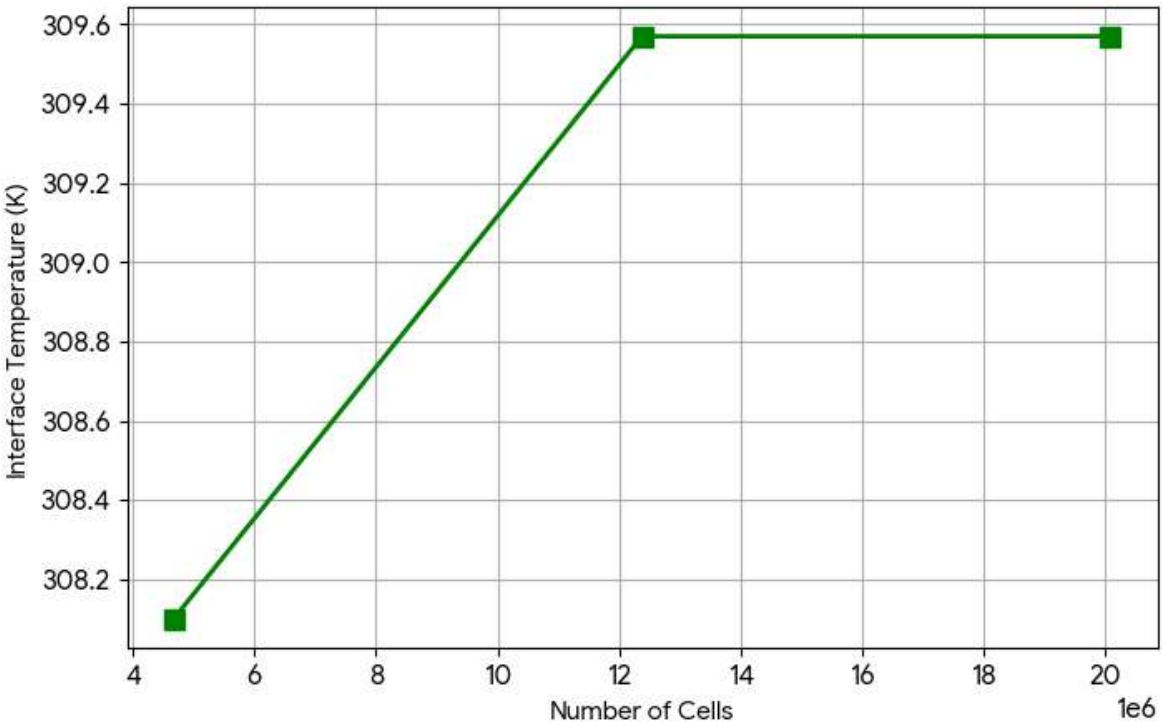


Figure 5.3, Grid independence study

5.3 WITH POROUS MEDIUM

To enhance the heat transfer performance of the cobweb-channel cold plate, a porous medium was incorporated into the fluid domain. The porous zone was implemented within the fluid section of the cobweb channels to increase the effective thermal conductivity and augment the convective heat transfer between the coolant and the cold plate structure. The porous medium selected for this study was a l aluminium, whose thermophysical and transport properties were assigned based on empirical data.

The porous region was characterized by a permeability (K) of 0.083×10^{-6} m² and a porosity (ε) of 0.95 from the below equation 5.1 ,allowing for the accurate prediction of pressure drop and flow resistance. The solid matrix of the metal was defined with a density of 2702 kg/m³ and a thermal conductivity of 237 W/(m·K), contributing significantly to the effective thermal conductivity of the composite region. Additionally, a specific heat of 903 J/(kg·K) was assigned to the porous solid material. The interfacial heat transfer between the fluid and the porous solid was modeled using an Interfacial Heat Transfer Coefficient

(IHTC) of 12345 W/(m²·K) and a surface area density of 1571 m⁻¹, ensuring a rigorous representation of the thermal non-equilibrium between the two phases.

$$K = \frac{\epsilon^2 d_p^2}{150(1-\varepsilon)^2} \tag{5.1}$$

$$A_{fs} = \frac{6(1-\varepsilon)}{d_p} \tag{5.2}$$

$$Nu = \frac{h_{fs} d_p}{k_f} = \begin{cases} 0.76 Re_d^{0.4}\, Pr^{0.37}\,, (1 \le Re_d \le 40 \\ 0.52 Re_d^{0.5}\, Pr^{0.37}\,, (40 \le Re_d \le 10^3 \\ 0.26 Re_d^{0.6}\, Pr^{0.37}\,, (10^3 \le Re_d \le 10^5 \end{cases} \tag{5.3}$$

The inclusion of the porous medium fundamentally alters the thermo-fluid behavior within the cold plate. While it increases the flow resistance compared to an open channel, the high surface area density and enhanced thermal conductivity of the metal significantly improve the dissipation of heat generated by the battery cells, thereby improving the overall thermal management efficiency of the cobweb-channel configuration.

CHAPTER 6: RESULTS AND DISCUSSION

This chapter presents the results of numerical simulations investigating the thermal and hydraulic performance of a cobweb channel coldplate for battery thermal management. Two configurations are compared: the baseline cobweb channel without porous media and the cobweb channel filled with porous media (pore diameter = 0.2 mm, porosity = 0.95). A parametric study was conducted by varying the inlet velocity from 0.025 m/s to 0.15 m/s while maintaining a constant battery volumetric heat generation of 32,357 W/m³. The key performance metrics evaluated include interface temperature, outlet fluid temperature, and pressure drop across the coldplate.

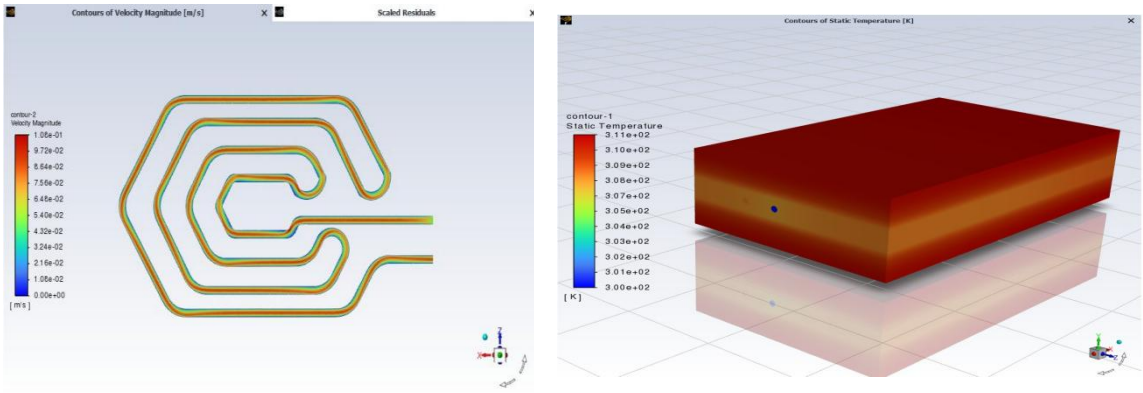


Figure 6.1, Contour: i) Velocity, ii) Static temperature

6.1 THERMAL PERFOMANCE ANALYSIS

6.1.1 Interface Temperature

The interface temperature between the battery and coldplate is the most critical parameter for battery thermal management, as it directly influences battery degradation, safety, and cycle life. Table no. 2 summarizes the interface temperatures obtained for both configurations across all velocities tested.

Table 6.1, Interface Temperature Comparison

Velocity (m/s)	Without Porous (K)	With Porous (K)	Temperature Reduction (K)
0.15	303.00	302.68	0.32
0.10	304.68	304.04	0.64
0.05	309.57	308.25	1.32
0.025	318.23	316.80	1.43

The addition of porous media consistently lowers the interface temperature across all velocities tested. The temperature reduction ranges from 0.32 K at the highest velocity (0.15 m/s) to 1.43 K at the lowest velocity (0.025 m/s). This indicates that porous media provides greater thermal benefit at lower flow rates, where the fluid residence time is longer, allowing more effective utilization of the increased surface area offered by the porous structure.The interface temperature decreases with increasing velocity for both configurations. At 0.15 m/s, the interface temperature reaches 302.68 K with porous media, representing the best cooling performance achieved in this study. This trend is attributed to the higher convective heat transfer coefficient at increased flow rates, which enhances heat removal from the battery interface.

6.1.2 Outlet Fluid Temperature and Thermal Efficiency

Table 6.2, Outlet Fluid Temperature Comparison

Velocity (m/s)	Without Porous (K)	With Porous (K)
0.15	302.68	302.60
0.10	304.02	304.02
0.05	308.02	308.02
0.025	316.00	316.00

A remarkable finding is that the outlet fluid temperature is nearly identical to the interface temperature in all cases, with a maximum deviation of less than 1 K. This demonstrates that the cobweb channel achieves near-complete thermal utilization, where the fluid exits at approximately the same temperature as the cooling surface. The thermal efficiency, defined as:

$$\eta = \frac{T_{out} - T_{in}}{T_{interface} - T_{in}} \tag{6.1}$$

exceeds 0.95 for all cases, confirming that the cobweb geometry alone provides sufficient surface area and residence time for the fluid to reach thermal equilibrium. This high efficiency persists regardless of whether porous media is present, suggesting that the cobweb geometry is inherently optimized for heat transfer.

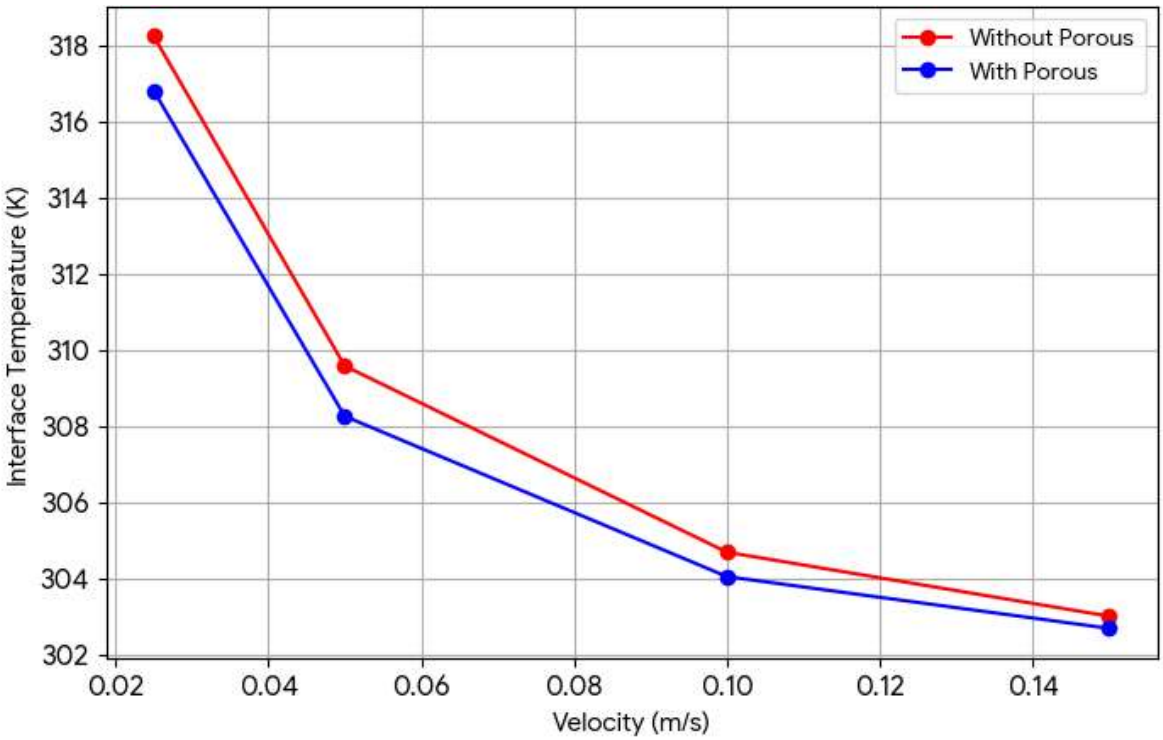


Figure 6.2, Interface Temperature vs Velocity

6.2 HYDRAULIC PERFOMANCE ANALYSIS

Table 6.3, Pressure Drop Comparison

Velocity (m/s)	Without Porous (Pa)	With Porous (Pa)	Pressure Ratio (With/Without)
0.15	703.33	2445.80	3.48
0.10	418.68	1604.95	3.83
0.05	149.55	792.04	5.30
0.025	81.63	394.52	4.83

The addition of porous media consistently increases pressure drop across all velocities tested, with pressure ratios ranging from 3.48 at 0.15 m/s to 4.83 at 0.025 m/s. This behavior is expected, as the porous structure introduces additional flow resistance through both viscous and inertial effects.

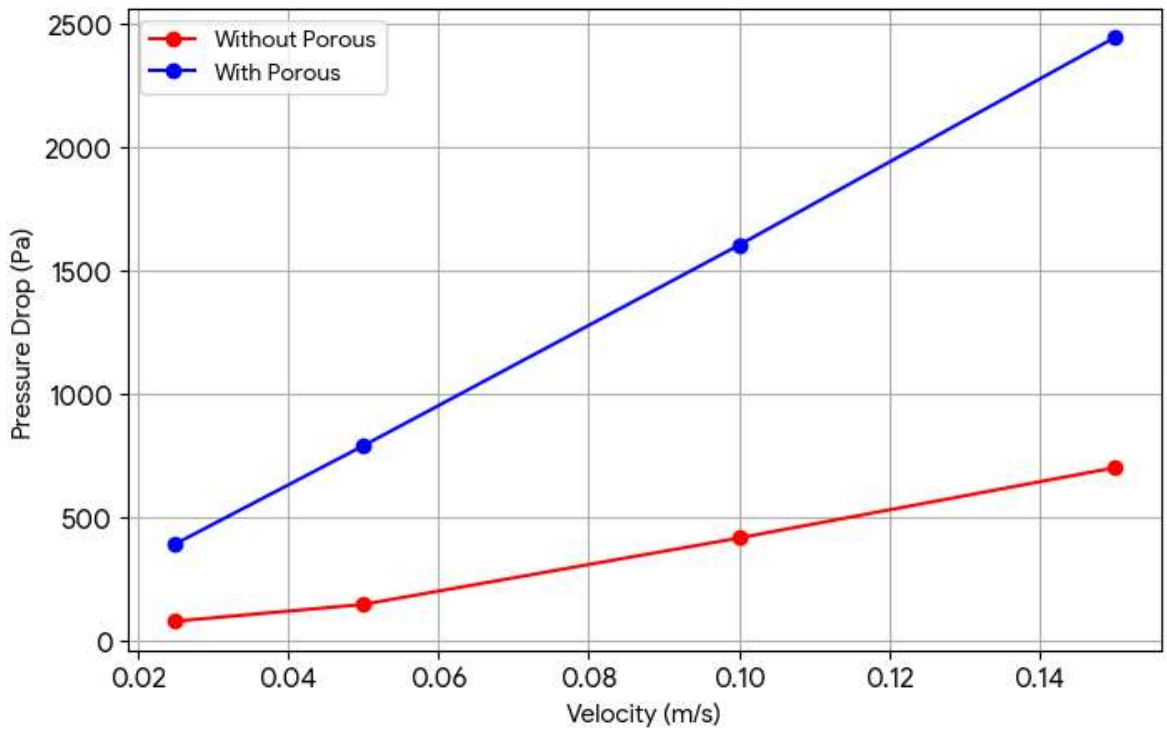


Figure 6.3, Pressure drop vs Velocity

6.3 PARAMETRIC STUDY

Table 6.4, Parametric study

Velocity (m/s)	With/Without	Volumetric heat (W/m³)	T interface (K)	Toutlet (K)	Pressure (Pa)
0.15	With	32357	302.68	302.6	2445.8
	Without	32357	303.0	302.68	703.33
0.1	With	32357	304.04	304.02	1604.95
	Without	32357	304.68	304.02	418.68
0.05	With	32357	308.25	308.02	792.04
	Without	32357	309.57	308.02	1499.55
0.025	With	32357	316.80	316.0	394.52
	Without	32357	318.23	316.0	81.63

6.3.1 Effect of Velocity on Thermal Performance

Increasing velocity from 0.025 m/s to 0.15 m/s reduces the interface temperature by approximately 15 K for both configurations. This demonstrates that flow rate is a dominant parameter controlling cooling performance. The relationship between velocity and interface temperature follows an approximately inverse trend.

6.3.2 Effect of Velocity on Pressure Drop

For the with-porous configuration, pressure drop increases with velocity, following an approximately quadratic relationship. For the without-porous configuration, pressure drop also increases with velocity, though at a lower rate. The pressure ratio (with/without) ranges from 3.48 to 4.83, indicating that the porous media contributes significantly to the overall flow resistance

CHAPTER 7: CONCLUSIONS

This study numerically investigated the thermal and hydraulic performance of a cobweb channel coldplate for battery thermal management, comparing a baseline configuration without porous media to a configuration fully filled with porous media (pore diameter = 0.2 mm, porosity = 0.95). A parametric study was conducted across velocities ranging from 0.025 m/s to 0.15 m/s.

Based on the results obtained, the following conclusions are the cobweb channel achieves near-ideal thermal efficiency across all velocities tested, confirming the effectiveness of the geometric design for battery cooling applications. The addition of porous media consistently lowers the interface temperature by 0.32 K to 1.43 K, with maximum benefit observed at the lowest velocity (0.025 m/s). Higher flow velocities (0.15 m/s) produce the lowest interface temperature (302.68 K with porous media), making them suitable for high-power applications where maximum cooling is required. The addition of porous media consistently increases pressure drop across all velocities, with pressure ratios ranging from 3.48 to 4.83. By this study we conclude as The optimal operating range for the cobweb channel with porous media is 0.10-0.15 m/s, balancing cooling performance (interface temperature ≤ 304 K) with acceptable pressure drop (≤ 1605-2446 Pa).

Based on the findings and limitations of this study, the following directions are recommended for future research. One promising direction for future work is the investigation of annular porous media insertion, where porous material is placed only near the channel walls rather than filling the entire channel cross-section. This approach focuses on adding porous media in the region where the temperature gradient is highest (near the heated wall).The annular configuration is expected to provide a favorable trade-off, offering meaningful thermal enhancement with significantly lower pressure drop penalty compared to full-channel filling.

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